Reference Material

Study of DS-RX1 System Realization (4 printers)

Aug. 31, 2017
Dai Nippon Printing Co., Ltd.

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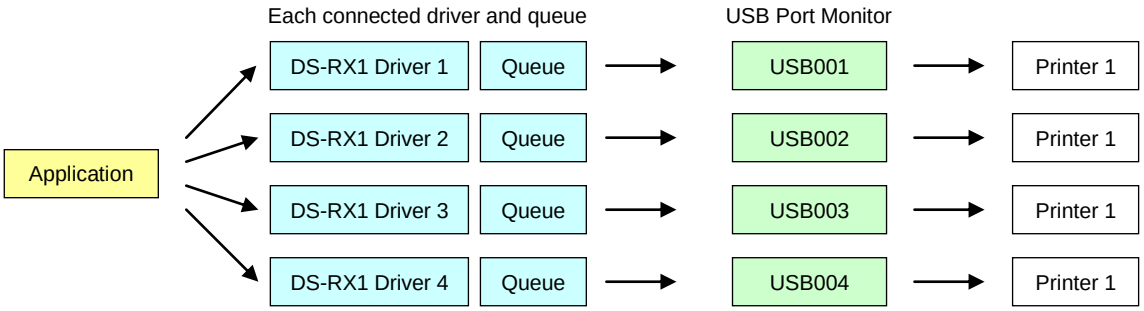
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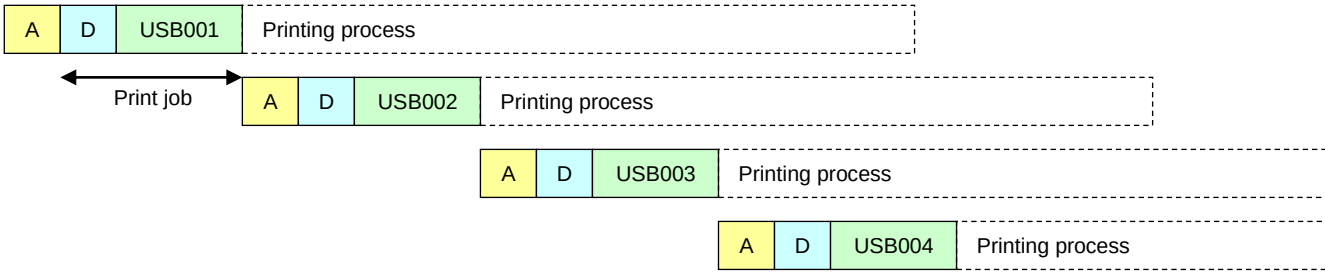
1. About the Data Flow on Windows

In the printer driver settings, there is a Spool On/Off setting. The data flow when using this setting is shown below.

1-1. Operation with Spool Off



When the application outputs data to each driver, the current driver can't release the print job until the data is output to the USB port. Therefore, the application waits until data output to the current driver's USB port is complete before it can output data to the next driver.

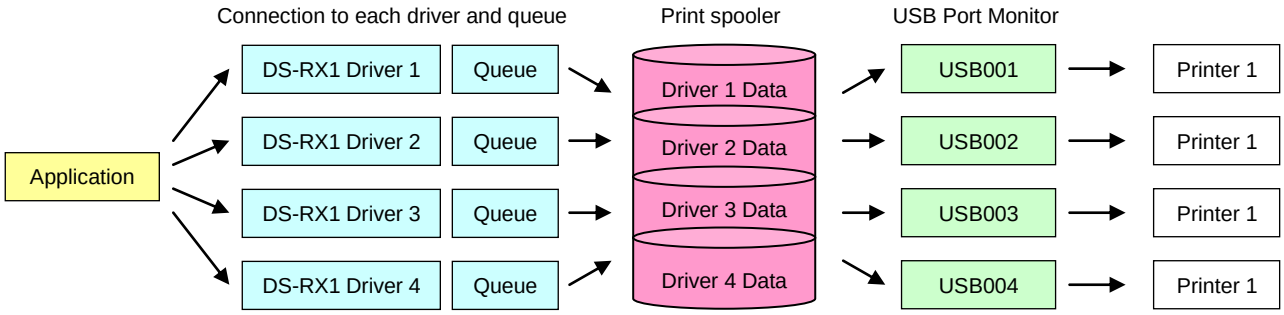


A	Application section: data resizing, status processing (API), sheet no. control, color management, color adjust, etc
D	Driver processing: color management, color adjust, printer command processing, etc.
USB	USB data transferring, data send/receive management

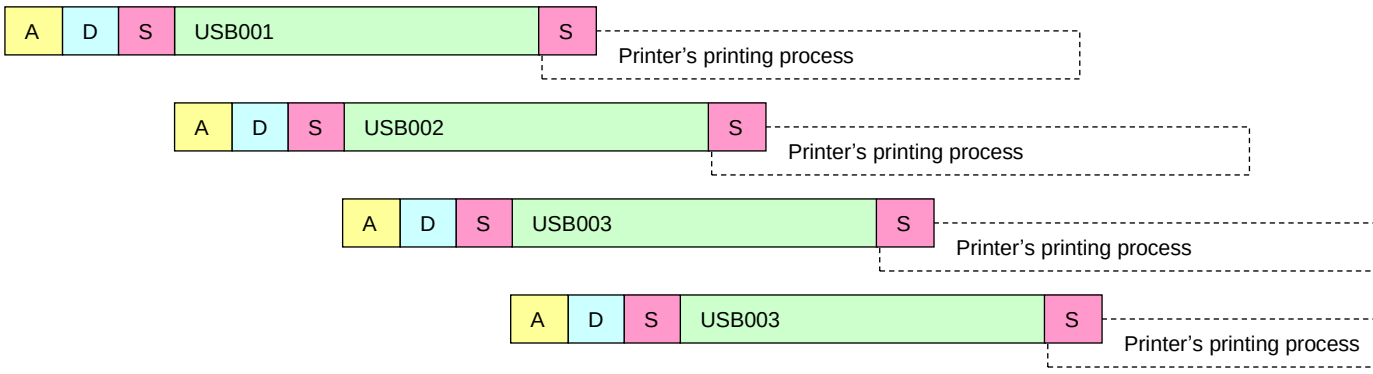
[About status check]

With Spool OFF, the application cannot perform other processes while sending data. Therefore, it cannot check printer status of other printers while it is sending data.

1-2. Operation with Spool On



When the application outputs the data to each driver, each driver outputs the data to the print spooler and releases the print job. The application can output data to the next driver without waiting for the data to be output to each USB port. When the print spooler takes in the data, USB Port Monitor control is performed for each port, and the data output to each port begins. If there are multiple data in the print spooler, data output to each USB port can be performed concurrently.



A	Application section: data resizing, status processing (API), sheet no. control, color management, color adjust, etc
D	Driver processing: color management, color adjust, printer command processing, etc.
S	File writing to and deletion from the print spooler
USB	USB data transferring, data send/receive management

[About status check]

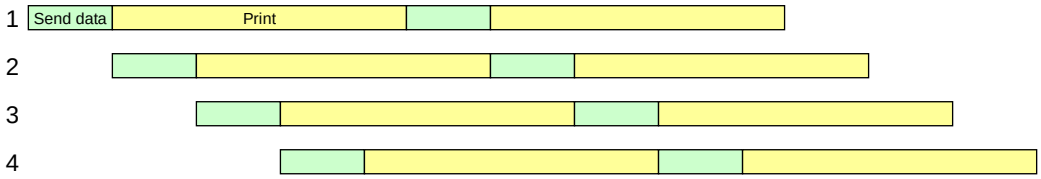
With Spool ON, once the data has been output to the spooler, the application is freed from the print processing and can perform other processes. Therefore, even while data is being sent, it is possible to check the status of other printers.

2. Relation between printer buffer operation (single/double) and Spool ON/OFF

The relationship between printer buffer operation (single/double) and Spool ON/OFF is shown below.

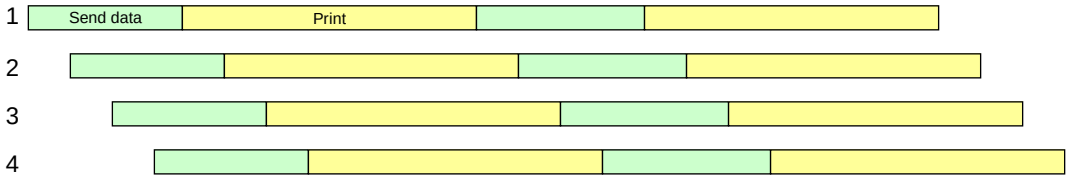
(1) Single buffer, Spool OFF

Because the spooler is OFF, it sends data to 1 printer at a time, and waits for the printing to be completed before sending the next data.



(2) Single buffer, Spool ON

By setting spool ON, send data processes overlap, but due to spool processing, the total processing time is longer.

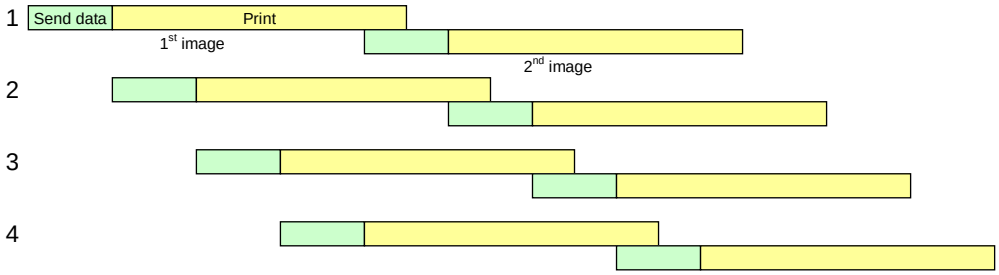


[Remarks]

- Theoretically, with Spool ON, the throughput time should be shorter, but with current hardware and operating systems, in sending data concurrently, the transmission time is longer, so we can surmise that it takes more time than with Spool OFF.
- With future improvements in hardware and operating systems, high speed transmission will be possible even when sending data concurrently, so Spool ON will likely be faster.

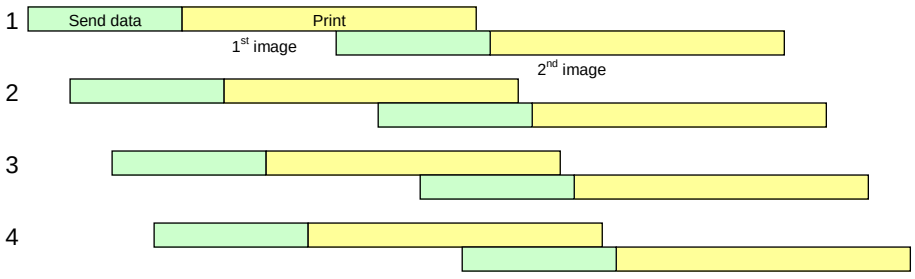
(3) Double buffer, Spool OFF

Using a double buffer, while the first image is being printed, data for the second can be sent, so processing time is that much shorter.



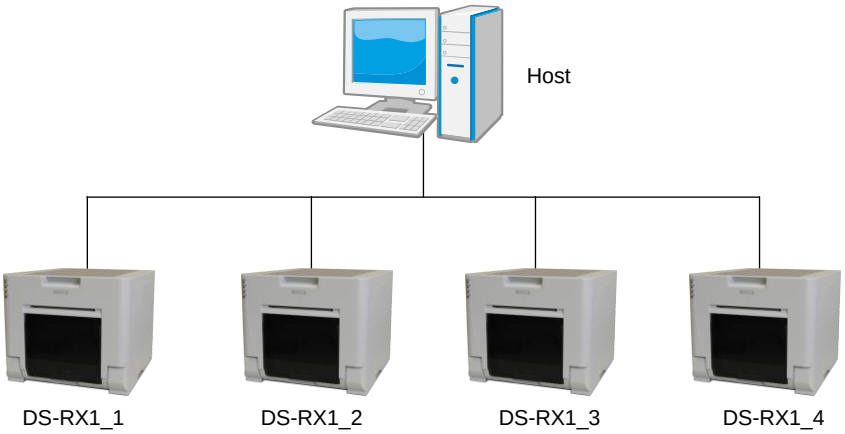
(4) Double buffer, Spool ON

With spool ON, data transmission can overlap, and with the double buffer process allowing data transmission during printing, the total processing time is the shortest possible.



3. System Composition

- Connected printers : 4 Unit
- Printer operating mode : Double buffer operation
- Spool setting : Spool ON



[Control Method]

The general process flow is that, when there is a print request, it looks for a printer to which data can be sent, and then sends the data. The sequence is as follows:

(1) When there is a print request, checks for print job at the current printer (Windows API EnumJobs). If there is a print job, changes the current printer to a different printer.
(When there is a print job, while the Status API tries to get the printer status, the application waits until there is no print job and it can get the status. In order to send data effectively, when there is a print job, it changes printers in order to find a free printer to which it can send data.)

(2) Performs error check. If there is an error, runs error recovery process (See "4. Error Processing")

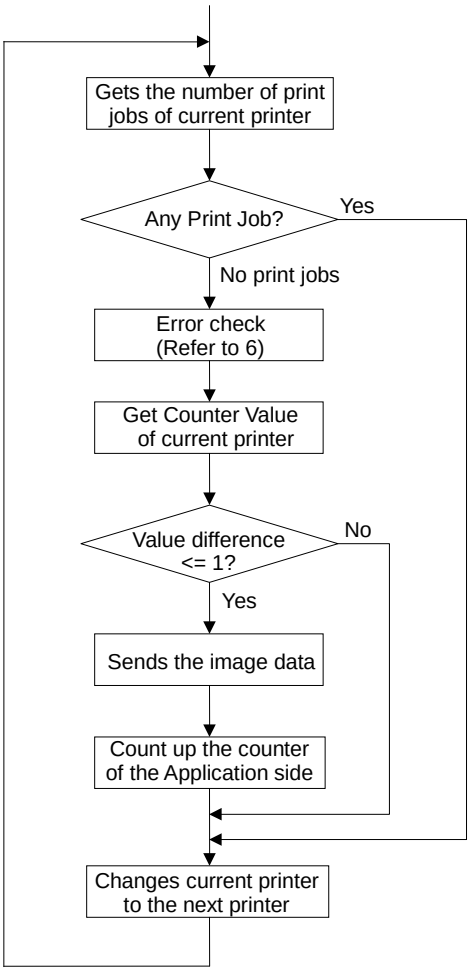
(3) When there are no print jobs or errors, it gets the printer count number (Status API GetCounterL).
It compares the application and printer count values to check for free buffers.

Count value difference = 0 --> 2 empty buffers
Count value difference = 1(*1) --> 1 free buffer
Count value difference = 2(*2) --> 0 free buffers

(4) If there is a free buffer, sends print data to that printer.

(5) After sending print data, counts up the application side count value and changes the current printer.

(6) If there are no free buffers, it changes the current printer and looks for a printer to which it can send data.



(*1) When printing 6x8(A5), 5x7(2L), 6x6(*4), 5x5(*4) or the dual image of 6x4 or 5x3.5(*3) size, the counter value difference will be 2.
(*2) When printing 6x8(A5), 5x7(2L), 6x6(*4), 5x5(*4) or the dual image of 6x4 or 5x3.5(*3) size, the counter value difference will be 4.

*3. Effective at firmware Ver 1.20 or later.
*4. Effective at firmware Ver 2.06 or later.

4. Error processing

When an error occurs, the printer count value is gotten as follows, and based on that value, it is determined how far printing proceeded. Then the affected data is resent to another printer.

(1) Checks for a print job, and if there is a print job, does not get status for that printer.
(2) If there is no print job, gets printer status, and checks for errors.
(3) If there is an error, gets the printer count value.
(4) Compares the printer and application count values.
(5) If the difference in count values is 2(*2), reprocesses the previous 2 data for printing (sends the data to another printer that is operating normally). If the difference in count values is 1(*1), reprocesses the previous 1 data for printing.

[Remarks]

- If there is a print job for too long, there may be a communication break, so error processing should take place.
- If there is no job, and for some reason there is a communication break, about 1 minute after DS-RX1 Printer Status API tries to get the status, a fail code will be returned. (During this time, the application will be waiting for status) When the program checks the status, it checks the time from start to finish, and if it is taking a long time, error processing is necessary.

